Power Wave 1

Single Phase, 2.1 to 17 kW

POWERFUL SINGLE PHASE SOLUTION

Looking for a high capacity, single enclosure, lighting inverter? The Power Wave 1 is our most compatible inverter to power almost all emergency lighting needs. A solution that is always on and always monitored. Power Wave 1 monitors not only utility power but also the internal system.

Scan QR Code

For the latest product news, technical specs, drawings, guides and manuals.

STANDARD FEATURES

- 90 Minute Backup Design at Full Load
- Listed to UL924 and UL1778 Standards
- Compatible with Self-Ballast Fluorescent, Incandescent, Halogen, Quartz Re-strike, HID, HPS and LED
- True Online, Double Conversion, Technology with **Zero Transfer Time**
- Pure Sine Wave
- TVSS Input Transient Voltage Surge Suppressor (Class C)
- High Frequency Pulse Width Modulation (PWM) for Conditioned Power to the Load
- Input Power Factor Correction and Full Voltage Regulation
- Multiple Output Voltages
- Microprocessor Controlled LCD with Visual Status and Event Indicators
- Self-Testing and Battery Exerciser with LCD Display
- DC Breakers for Ease of Maintenance

OPTIONAL FEATURES

- Global Monitoring (GMS):
  - GMS 1: Local event Logger with Memory (PC Connections via RS232 and RS485) 500 Events
  - GMS 2: Remote Monitoring with Local Network Connection (WiFi, Ethernet or Modem)
  - GMS 3: Remote Monitoring with Cellular Network Connection
- Remote Status Panel
- Battery Monitoring (String or Per Battery)
- Dry Contacts Normally Open
- Form C Dry Contact with N/O or N/C Field Selectable for Desired Signal
- Main Input & Output Circuit Breaker (UL489)
- Normally ON and/or Normally OFF Auxiliary Breakers (24 MAX)
- Custom Rated KAIC Breakers
- 65 KAIC Total System Short Circuit Current Rating
- Internal or External Maintenance Bypass Switch
- Seismic Bracket
- High Temperature Batteries
- Long Life Battery 20 Year Warranty
- Time Delayed Transfer to Battery
- Time Delayed Transfer from Battery
- Maintenance:
  - Extended Warranty
  - Preventative Maintenance Plans

ISO9001 Certified Since 1996

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110-DS-10-22

NFPA CODES & STANDARDS COMPLIANT

MADE IN USA

SPECIFICATIONS

Power Wave 1

Single Phase
2.1 to 17 kW

*All units are 90 minutes battery backup time @ full load.
Consult factory for backup times (up to 120 minutes)

SAFETY STANDARDS
UL924, UL1778, NFPA101, NFPA70, NEMA, and ANSI
Listed to UL924 and UL1778 by Standards

INPUT
AC Voltage: -15% / +10%
Frequency: Slew rate of 0.2 Hz/sec

OUTPUT
AC Voltage Regulation: ±5%,
2 wires plus-ground for single voltage outputs,
3 wires plus-ground for dual voltage outputs
Wave Form: Sinusoidal
Harmonic Distortion: < 5% THD; < 3% Single Harmonic

PROTECTION
Overload: 115% for 5 - 10 min, 125% for 30 sec

BATTERY
Battery: Sealed, maintenance free VRLA lead acid battery
Recharge Time: Conforms to UL standards

NOISE ISOLATION
Common Mode: -120 dB
Transverse Mode: -60 dB

ENVIRONMENT
Operating Temp: 0° to 40°C (32° - 104°F)
Battery: 20° to 25°C (68° - 77°F)
Electronic Storage Temp: -20° to 70°C. (-4° - 158°F)
Humidity: 0% to 95% (non-condensation)
Altitude: Up to 10,000 ft

Cabinet Dimensions

(Inches) Width x Height x Depth

Cabinet A
39"W x 48"H x 18"D

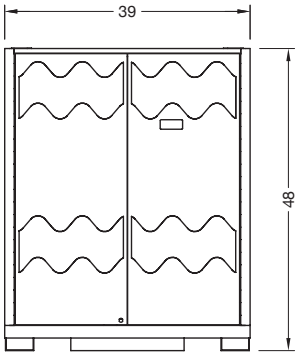
Cabinet B
39"W x 68"H x 18"D

Cabinet C
51"W x 70"H x 30.5"D

*120min 17kW require two 'C' cabinets.

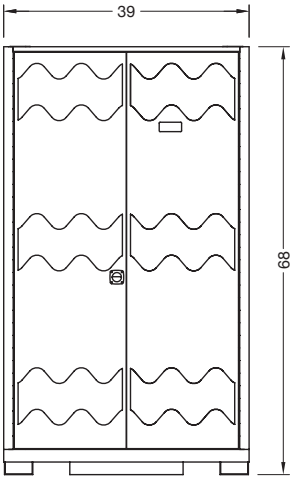
Cabinet A

90min: 2.1-3kW



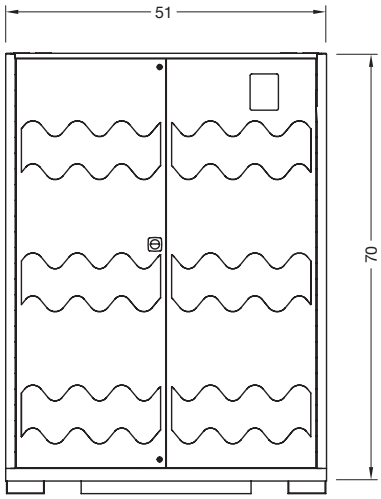
Cabinet B

90min: 5 to 8kW
120min: 3 to 5kW



Cabinet C

90min: 10 to 17kW
***120min:** 5 to 15kW



Model Numbers

Model number will be reflected on name plate, not designed for ordering or specifications.

Unit Name	Capacity (kW)		Input (Volts)		Output (Volts)	
PW Power Wave 1	**2.1**	2.1kW	**A**	120V	**0100N1**	120V
	3.0	3kW	**B**	208V	**1300N1**	208V
	5.0	5kW	**D**	240V	**0400N1**	240V
	6.0	6kW	**R**	277V	**2500N1**	277V
	8.0	8kW	**H**	480V	***5800T1**	120V/240V
	010	10kW				120V/277V
	012	12.5kW				277V/480V
	015	15kW				120V/240V/277V
	017	17kW				

*5800T1 part number is reserved for dual voltage and multi voltage outputs.
We recommend 120V/240V for any 120V output needs. Provides the ability to feed two hots to a panel.

BTU / Weight

	*Weight (LBs)	
Capacity (kW)	**90min**	**120min**
2.1	826	982
3.0	1066	1020
5.0	1284	1524
6.0	1284	1596
8.0	1464	1833
10	2870	2913
12.5	3777	3657
15	4512	5300
17	4512	5800

Approximately 1000 BTU's per hour for every 3 kilowatts. The approximation is worst case BTU output, measured during recharge following a discharge.

Warranty

Standard warranty is eighteen (18) months from ship date or twelve (12) months from start up, whichever occurs first. Optional second year warranty with factory performed preventative maintenance available.

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NFPA CODES & STANDARDS COMPLIANT
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ORDERING GUIDE

PROJECT NAME :

REP / DISTRIBUTOR :

ORDER NUMBER **DOES NOT** REFLECT MODEL NUMBER. PLEASE CONSULT FACTORY.

Power Wave 1

Example: PW-3.0-A02-S90-N08A40-FT-FC-GMS3-PM2

ELECTRICAL DESIGNER / ENGINEER :

DATE :

PW-

PW — MODEL SERIES

PW: Power Wave 1

3.0 — KW RATING

2.1: 2.1kW
3.0: 3kW
5.0: 5kW
6.0: 6kW
8.0: 8kW
010: 10kW
012: 12.5kW
015: 15kW
017: 17kW

A02 — VOLTAGE INPUT/OUTPUT

INPUT	OUTPUT
* **A** - 120V	**01** - 120V
B - 208V	**02** - 208V
D - 240V	**03** - 240V
R - 277V	**04** - 277V
H - 480V	**05** - 120V / 240V
	06 - 277V / 480V
	07 - 120V / 277V
	** **08** - 120V / 208V / 277V

* 120V input only available up to 8 kW output.
** Consult factory for 3 voltage output systems.

S90 — BATTERY TYPE

S90: Standard @ 90min
* **L90:** Long Life @ 90min
H90: Hi Temp @ 90min
U90: USA @ 90min
S120: Standard @ 120min
* **L120:** Long Life @ 120min
H120: Hi Temp @ 120min
U120: USA @ 120min

* Long Life 20 Yr. Pro Rata Battery.

N08A40 — AUX BREAKERS (UL1077)

AUX BREAKERS EXAMPLE

N 08 A 40

OUTPUT
N: Normally ON
F: Normally OFF
****FD:** Normally OFF w/ Time Delay

BREAKERS
***01 - 24** (Max 24)

VOLTAGE
A - 120V, 1P
B - 208V, 2P
C - 240V, 2P
D - 277V, 1P
E - 480V, 2P

****AMP RATING**
10, 15, 20, 25, 30, 40, 50, 60

* Only up to 24 poles.
** Allows you to have individual circuits remain on battery even when utility is restored for a period of time. Must be normally a OFF circuit.
***Consult factory for custom amp rating.

FT + FC + GMS3 — OPTIONS

PICK ANY:
X: Seismic Mounting
MB: Internal Maintenance Bypass
EB: Ex. Maintenance Wraparound Bypass Switch
SCCR: 65 KAIC Total System Short Circuit Current Rating
IB: Input Breaker (UL 489)
OB: Output Breaker (UL 489)
DS: Drip Shield Hood
CL: Corbin Lock

BATTERY OPTIONS
FC: Fast Charge (12 hrs or less)
TR: Thermal Runaway Gassing Control without Battery Disconnect (IFC 1206.2)

MONITORING OPTIONS
C: Normally Open Dry Contacts
DCT2: Dual, Open and Closed Dry Contacts
A: Local Audible Alarm with Silencer Switch
CP: RS 232 and RS485 Connecting Port
P: Remote Status Panel (Hard Wired with Extended Cable)

SYSTEM TRANSFER
FT: Fast Transfer with 2 Milliseconds
TD(B): User Selectable Time Delay to Battery for Capacitor Discharge During EM (45 Milliseconds Standard)
TD(U): User Selectable Time Delay Back to Utility During Power Restoration for Facility Inrush

BATTERY MONITORING SYSTEM (BMS)
PICK ONE:
SM: String Monitoring
BM: Battery Monitoring
BM-TS: Temperature Sensor

GLOBAL MONITORING SYSTEM (GMS)
PICK ONE:
GMS 1: Local Accessible Event Logger with Unit Audible Alarm
GMS 2: Testing Logs with Remote Accessibility Through Local Network
GMS 3: Testing Logs with Remote Accessibility through Local Network and GPRS for Cellular Text Alerts

PM2 — WARRANTY & SERVICES

PICK ANY:
Blank: Standard Onsite Startup and Warranty

ADDITIONAL:
EXT: Additional One Year Extended Warranty (Includes First Year Preventative Maintenance)
PM (2-3): Onsite Preventative Maintenance (Once Per Year, up to 3 Years)
T1: 4 Hour Training at Time of Start Up
T2: Additional Training day up to 8 hours

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